

• **NODE-ATTRIBUTED GRAPH** → enrich nodes with contextual / semantic info

- node type
- timestamp
- text / documents
- numerical values
- images
- categorical values

introduce a set of functions $\{f_1, \dots, f_{d_v}\}$ s.t. $f_i: E \rightarrow S_i$
generic set of values $S_i = \mathbb{R}$ assign to e a vector $\vec{u} \in \mathbb{R}^{d_v}$

$[u_1, \dots, u_{d_v}]$
stacking $\vec{u}$ for all links $\Rightarrow U \in \mathbb{R}^{m \times d_v}$
NODE FEATURE MATRIX

attributes a set of functions easily fit into associative memory data structures, s.a. dictionary, maps

$$\begin{aligned} \{f_1: E \rightarrow S_1, \dots, f_d: E \rightarrow S_d\} \\ \theta = (u, v) = \{ \begin{aligned} &"f_1": \langle \text{value in } S_1 \rangle, \\ &\vdots \\ &"f_m": \langle \text{value in } S_d \rangle \end{aligned} \end{aligned}$$

function set Tuple e.g. $(u, v)[f_1] = 4.75$

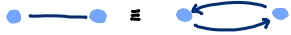
• **REPRESENTING GRAPH**

Δ **ADJACENCY MATRIX** → inspired by $E \subseteq V^2$

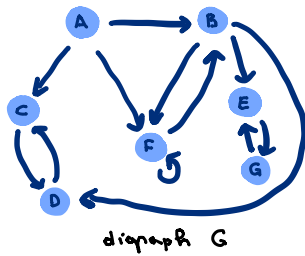
$$A \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|} \text{ s.t. } A_{u,v} = 1 \Leftrightarrow (u, v) \in E$$

squared matrix

• for undir. graph A is symmetric $A = A^T$
 $(u, v) \in E \Leftrightarrow (v, u) \in E$



• for dir. graph the symmetry holds for reciprocal links only, e.g. $c \rightarrow d$ and $d \rightarrow c$



	A	B	C	D	E	F	G	
A	0	1	1	0	0	1	0	3
B	0	0	0	1	1	1	0	3
C	0	0	0	1	0	0	0	1
D	0	0	1	0	0	0	0	1
E	0	0	0	0	0	0	1	1
F	0	1	0	0	0	1	0	2
G	0	0	0	0	1	0	0	1
	0	2	2	2	2	3	1	12

$\sum_{v=1}^{|\mathcal{V}|} A_{u,v}$ row-wise summation

$\sum_{u=1}^{|\mathcal{V}|} A_{u,v}$ column-wise summation

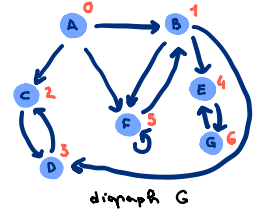
A is good for dense graph but waste of memory to store "0": K = bytes for a single element $\Rightarrow m^2 K$ for A

$\mathcal{O}(K)$ for useful info in sparse graph $\mathcal{O}(m) \approx m \cdot K$

if $K=8$ (double in C) $\rightarrow 7^2 \cdot 8 = 392$
 $12 \cdot 2 \cdot 8 = 196$
store (u, v) = position of "1"s

format for SPARSE (ADJACENCY) MATRIX

• **COORDINATE LIST - COO**
set of entry (row, column, value)



A	A	B	B	B	C	D	E	F	F	G
B	C	F	F	D	E	D	C	G	F	B
1	1	1	1	1	1	1	1	1	1	1

0	0	1	1	2	3	4	5	5	6
1	2	5	5	3	4	3	2	6	5
1	1	1	1	1	1	1	1	1	1

format in $P_1 G$ (attribute edge-index)

• **Compressed Sparse Row - CSR**
3 indexed structures row $\in \mathbb{R}^m$

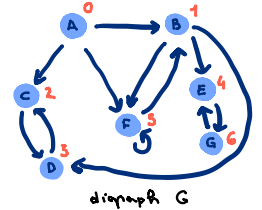
$$[i_1, \dots, i_m]$$

index for mode at j indicating where the set of modes incident to j starts

$$\text{col_ind} \in \mathbb{R}^e \text{ indexed set of links } [\dots, v_1, \dots, v_K, \dots]$$

$(u, v_1), \dots, (u, v_K)$

$$\text{val} \in \mathbb{R}^e \text{ weight or attribute functions for each link}$$



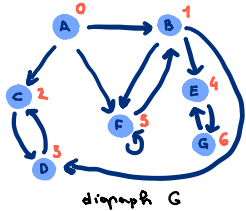
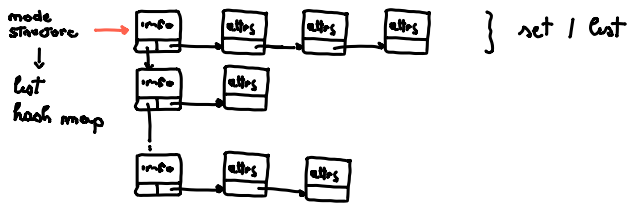
$$\text{col_ind} = [1, 2, 5, 3, 4, 5, 3, 2, 6, 1, 5, 4]$$

$$\text{row} = [0, 3, 6, 7, 8, 9, 11]$$

$$\text{mode} = [A, B, C, D, E, F, G] \rightsquigarrow \begin{matrix} A \rightarrow 0 & C \rightarrow 2 & E \rightarrow 4 \\ B \rightarrow 1 & D \rightarrow 3 & F \rightarrow 5 \end{matrix}$$

Δ **ADJACENCY LIST** : more dynamic data structure w.r.t. CSR but same idea = store adjacent modes of u

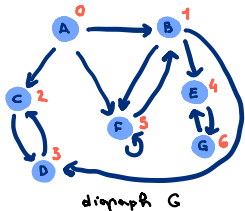
pointers instead of integer index



- $A \rightarrow \{B, C, F\}$
- $B \rightarrow \{D, E, F\}$
- $C \rightarrow \{D\}$
- $D \rightarrow \{C\}$
- $E \rightarrow \{G\}$
- $F \rightarrow \{B, E\}$
- $G \rightarrow \{E\}$

Δ **EDGE LIST** : based on idea graph = set of links

list of tuples $(u, v, \text{weight?})$

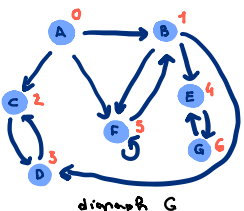


- $(A, B), (A, C), (A, F), (B, D), (B, E), (B, F),$
- $(C, D), (D, C), (E, G), (F, F), (F, B), (G, E)$

Δ **INCIDENT MATRIX** : $A_I \in \mathbb{R}^{m \times e}$
rectangular matrix

$$(u, v) \in E \Leftrightarrow A_I(u, (u, v)) = 1 \wedge A_I(v, (u, v)) = -1$$
$$\{u, v\} \in E \Leftrightarrow A_I(u, (u, v)) = 1 \wedge A_I(v, (u, v)) = 1$$

else 0 → arise with self-loop
else 0



- $e_1 (A, B), e_2 (A, C), e_3 (A, F), e_4 (B, D), e_5 (B, E), e_6 (B, F),$
- $e_7 (C, D), e_8 (D, C), e_9 (E, G), e_{10} (F, F), e_{11} (F, B), e_{12} (G, E)$

	e_1	e_2	e_3	e_4	e_5	e_6	e_7	e_8	e_9	e_{10}	e_{11}	e_{12}
A	1	1	1	0	0	0	0	0	0	0	0	0
B	-1	0	0	1	1	1	0	0	0	0	-1	0
C	0	-1	0	0	0	0	1	-1	0	0	0	0
D	0	0	0	-1	0	0	-1	1	0	0	0	0
E	0	0	0	0	-1	0	0	0	1	0	0	-1
F	0	0	-1	0	0	-1	0	0	0	1	1	0
G	0	0	0	0	0	0	0	0	-1	0	0	1

• in $E(G)$ elements are 2-elements sets $\{u, v\}$ but connections may bind together more than 2 nodes at the same time

remove the constraint and allow $E = \{v_1, \dots, v_k\}$
 $k \geq 2$

HYPEREDGE
 ↓
 HYPERGRAPH



DIRECTED HYPERGRAPH

$$H = (V, E)$$

$$E \subseteq 2^V \times 2^V$$

power set of V = set of all subsets of V

$$V = \{v_1, v_2, v_3\} \quad 2^V = \{\{v_1\}, \{v_2\}, \{v_3\}, \{v_1, v_2\}, \{v_1, v_3\}, \{v_2, v_3\}, \{v_1, v_2, v_3\}, \emptyset\}$$

on element $e \in E$ (U, W) with $U = \{v_1, v_2, \dots, v_s\}$ and $W = \{v_1, v_2, v_3\}$

TAIL

HEAD

(not a mistake in KG terminology is the opposite)
 in HG is tail $\rightarrow$ head (very intuitive :))

$\rightarrow$ again the approach is based on cartesian product



UNDIRECTED HYPERGRAPH

move from $(u, w) \rightarrow \{u, w\} \Rightarrow (U, W) \rightarrow \{U, W\} = U \cup W \in 2^V$

$$\rightarrow E \subseteq 2^V$$

Tool: hypergraphx.readthedocs.io HGX (Python)

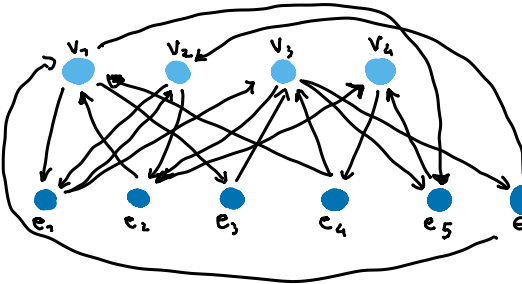
Q_1 : density for dir. hypergraph from $m = |V|$ and $l = |E|$
 for undir. hypergraph

H represented by incidence structures where $V_I = V \cup E$ and $E_I = \{(u, e) \mid u \in V, e \in E, u \in \text{tail}(e)\} \cup \{(e, u) \mid u \in V, e \in E, u \in \text{head}(e)\}$

by definition $V \cap E = \emptyset \Rightarrow V, E$ are partitioning V_I

directed structure

$$\begin{aligned} V &= \{v_1, v_2, v_3, v_4\} \\ E &= \{e_1 = (\{v_1, v_2\}, \{v_2, v_3\}), \\ &\quad e_2 = (\{v_2, v_3, v_4\}, \{v_1, v_4\}), \\ &\quad e_3 = (\{v_1\}, \{v_3\}), \\ &\quad e_4 = (\{v_4\}, \{v_3, v_1\}), \\ &\quad e_5 = (\{v_3, v_1\}, \{v_4\}), \\ &\quad e_6 = (\{v_1\}, \{v_2, v_1\})\} \end{aligned}$$



	e_1	e_2	e_3	e_4	e_5	e_6
v_1	1, 0	0, 1	1, 0	0, 1	1, 0	0, 1
v_2	1, 1	1, 0	0, 0	0, 0	0, 0	0, 1
v_3	0, 1	1, 0	0, 1	0, 1	1, 0	1, 0
v_4	0, 0	1, 1	0, 0	1, 0	0, 1	0, 0

membership of vertices in sets as in hypergraphs can be represented as BIPARTITE GRAPH or TWO-MODE NETWORK

two types of vertices +
 CONSTRAINT: edges run only between vertices of different types

make sense to use incidence matrix $B_I \in \mathbb{R}^{|V| \times |C|}$
 $B_{u,c} = \begin{cases} 1 & \text{if } u \in c \\ 0 & \text{else} \end{cases}$



Sometimes more convenient to work of a set of mode types

ONE-MODE PROJECTION

- select V on C as $V(G_P)$
- $(u, v) \in E(G_P) \Leftrightarrow u, v \in V(G_P) \quad \exists c \in V(B) - V(G_P) \text{ s.t. } (u, c), (c, v) \in E(B)$

projection discards a lot
 of info

• # groups shared by u and v

add weight function $w: E(G_P) \rightarrow \mathbb{R}$

$$w((u, v)) = |\{c \in V(B) - V(G_P) \mid (u, c), (c, v) \in E(B)\}|$$

$\approx$ # modes of the other type in common between u and v



each 'c' generates a clique of nodes
 in the mode-projection set

union of clique

in matrix notation

$$B = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 & 1 \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \end{bmatrix}$$

$$AND: B_u: \odot B_v: \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \end{bmatrix}$$

$$\downarrow \text{sum} \quad \text{③}$$

$$\begin{aligned} P_{u,v} &= \sum_{c \in V(B) - V(G_P)} B_{u,c} B_{v,c} \\ &= \sum_{c \in V(B) - V(G_P)} B_{u,c} B_{c,v}^T \\ &= B_u: B_v^T \end{aligned}$$

$$P = B B^T + \text{set diagonal to zeros}$$

Since we can represent hypergraph (undir or dir) we can project hypergraphs into graphs by one-mode projection

- most systems are made by interacting networks $\rightarrow$ # type of modes
 $\rightarrow$ # type of relationships with same types of vertices
 with # types of endpoints

M layers = set of M networks representing interactions within a layer

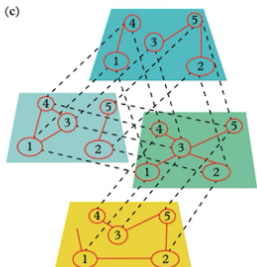
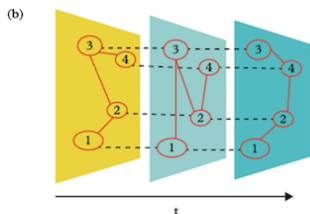
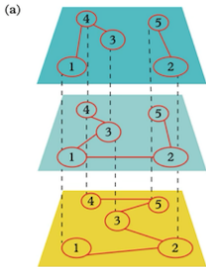
$M(M-1)$ layers = interactions between modes in pairs of # layers

no agreement on terminology
 for types of multilayer networks

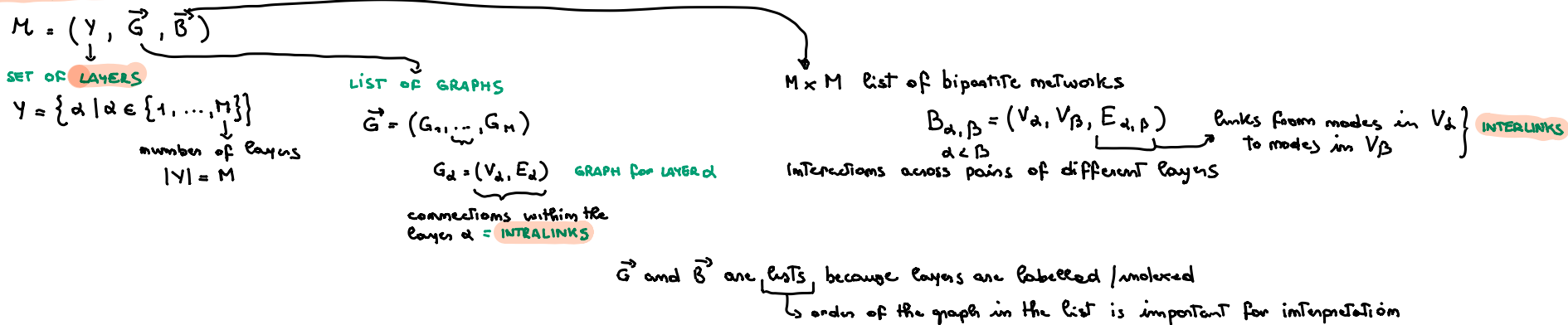
MULTIPLEX

MULTI-SLICE

NETWORK OF NETWORKS

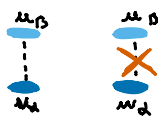


● GENERAL MULTILAYER NETWORK



● MULTIPLEX NETWORK

- simplifies multilayer definition
- one-to-one mapping of modes in different layers = **REPLICA NODES**
 - interlinks can EXCLUSIVELY connect the corresponding replica modes



one-to-one property
 ↳ interactions between the same set of modes
 AND each layer is a type of relationship

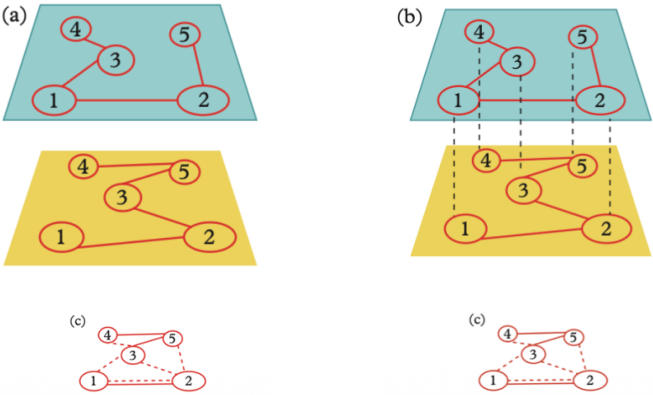
MULTIPLEX WITHOUT INTERLINKS

no distinction between the identity of replica modes → NO INTERLINKS
 expliciting the one-to-one mapping
 ⇒ COLLAPSE OF REPLICA NODES

MULTIPLEX WITH INTERLINKS

distinction on the identity of replica modes
 e.g. transportation networks
 ↳ interlinks = cost of modality change

$M = (\gamma, \vec{G}, \vec{B})$
 ↓
 for each $G_\alpha = (V_\alpha, E_\alpha)$ s.t.
 $V_\alpha = V = \{v_i \mid i \in \{1, \dots, m\}\}$



$V \rightarrow V_\alpha = \{(i, \alpha) \mid i \in \{1, \dots, m\}\}$
 ↓
 labeling encodes one-to-one mapping

$M = (\gamma, \vec{G}, \vec{B})$
 $G_\alpha = A^\alpha \in \mathbb{R}^{m \times m}$
 $B_{\alpha, \beta}$ s.t.
 $E_{\alpha, \beta} = \{((i, \alpha), (i, \beta)) \mid i \in \{1, \dots, m\}\}$

full info encoded in M adjacency matrices

A^α for the layer α , i.e.

$A_{u,v}^\alpha = 1 \Leftrightarrow$ in the layer α
 u and $v \in V$ are connected

► extensible for undir | dir and weighted | unweighted

► single graph with multiple links = **MULTI (DI) GRAPH**

↓
 E is a multiset of 2-mode sets

multiset allows multiple links



put a color = LABEL on links
 ↳ introduce $L \rightarrow G = (V, E, L)$
 ↳ label set, e.g. $\{1, \dots, \ell\}$
 ↳ $E \subseteq V \times V \times L$

► useful for dynamical processes

single representation in matrix form for M
SUPRA-ADJACENCY MATRIX
 $A \in m \cdot M \times m \cdot M$, $A_{(i, \alpha), (j, \beta)} = \begin{cases} A_{ij}^\alpha & \alpha = \beta \\ \delta(i, j) & \alpha \neq \beta \end{cases}$
 ↳ Kronecker delta 1 if $i=j$

$A = \begin{pmatrix} A^1 & I & I & I \\ I & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ I & I & \cdot & A^M \end{pmatrix}$

where $I \in \mathbb{R}^{m \times m}$
 is the IDENTITY MATRIX

► AGGREGATED NETWORK

= graph resulting from the interactions neglecting the type $\tilde{G} = (V, \tilde{E})$

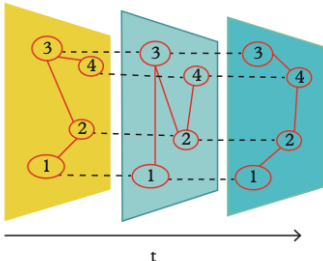
$\tilde{A}_{u,v} = \begin{cases} 1 & \text{if } \sum_{\alpha=1}^M A_{u,v}^\alpha > 0 \\ 0 & \text{else} \end{cases}$

● MULTI-SLICE NETWORK

special type of multilayer graph

- one-to-one mapping of modes in different layers = **REPLICA NODES**
- interlinks can EXCLUSIVELY connect replica modes of pairs of layers in temporal succession

interlinks in multislice connect each node in one layer only to the same mode in the next layer



γ and $\vec{G}$ as in the previous formalization
 AND
 $((i, \alpha), (i, \alpha+1))$
 successive layers at time $\alpha+1$

$\Rightarrow E_{\alpha, \alpha+1} = \{((i, \alpha), (i, \alpha+1)) \mid i \in \{1, \dots, m\}\}$

⇓

$A = \begin{pmatrix} a^{[1]} & I & 0 & \dots & 0 & 0 \\ 0 & a^{[2]} & I & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & a^{[M-1]} & I \\ 0 & 0 & 0 & \dots & 0 & a^{[M]} \end{pmatrix}$

● general formalism based on supra-adjacency matrix

every mode of α is $(u, \alpha) \rightarrow V_\alpha = \{(u, \alpha) \mid u \in \{1, \dots, m_\alpha\}\}$

↳ each layer has its own number of modes

$G_\alpha \equiv A^\alpha \in \mathbb{R}^{m_\alpha \times m_\alpha}$

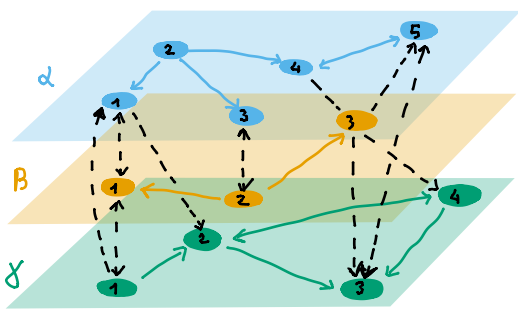
$B_{\alpha, \beta} \in \mathbb{R}^{m_\alpha \times m_\beta}$ s.t. $\beta_{u,v}^{\alpha, \beta} = \begin{cases} 1 & (u, \alpha) \rightarrow (v, \beta) \\ 0 & \text{else} \end{cases}$

special case of supra-matrices

$A = \begin{pmatrix} A^{1,1} & B^{1,2} & \dots & B^{1,M} \\ B^{2,1} & A^{2,2} & \dots & B^{2,M} \\ \vdots & \vdots & \ddots & \vdots \\ B^{M,1} & \dots & \dots & A^{M,M} \end{pmatrix}$

$A \in \mathbb{R}^{\hat{m} \times \hat{m}}$

$\hat{m} = \sum_{\alpha=1}^M m_\alpha$



$$A = \begin{array}{c|cccc|ccccc|ccccc} & 1,\alpha & 2,\alpha & 3,\alpha & 4,\alpha & 5,\alpha & 1,\beta & 2,\beta & 3,\beta & 1,\gamma & 2,\gamma & 3,\gamma & 4,\gamma \\ \hline 1,\alpha & & & & & & & & & & & & \\ 2,\alpha & & & & & & & & & & & & \\ 3,\alpha & & & & & & & & & & & & \\ 4,\alpha & & & & & & & & & & & & \\ 5,\alpha & & & & & & & & & & & & \\ \hline 1,\beta & & & & & & & & & & & & \\ 2,\beta & & & & & & & & & & & & \\ 3,\beta & & & & & & & & & & & & \\ \hline 1,\gamma & & & & & & & & & & & & \\ 2,\gamma & & & & & & & & & & & & \\ 3,\gamma & & & & & & & & & & & & \\ 4,\gamma & & & & & & & & & & & & \end{array}$$

all remaining entries are 0's

2 NETWORK OF NETWORKS → multi-layer networks with a supernetwork

$$G_L = (V_L, E_L)$$

$$V_L = \{\alpha | \alpha \in \{1, \dots, M\}\}$$

$E_L \approx$ connections between layers

$$L_{\alpha,\beta} = \begin{cases} 1 & \text{layer } \alpha \text{ is connected to layer } \beta \\ 0 & \text{otherwise} \end{cases}$$

constraint on mode connectivity only $((u, \alpha), (v, \beta))$ s.t. $L_{\alpha,\beta} = 1$

WITH REPLICA NODES : interlinks connect replica nodes

only if $L_{\alpha,\beta} = 1$

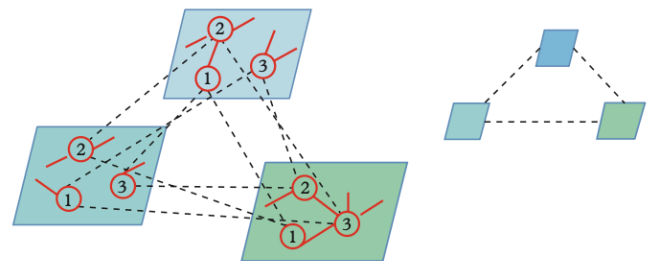
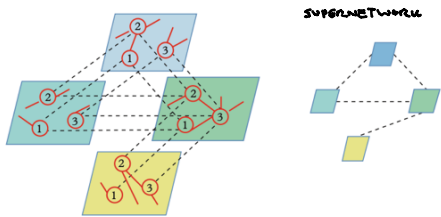
$$\beta_{u,v}^{\alpha,\beta} = L_{\alpha,\beta} \cdot \delta(u,v) \rightarrow \text{identity } I \text{ if } \alpha \text{ and } \beta \text{ layers are connected}$$

WITHOUT REPLICA NODES

interlinks only connect nodes belonging to layers connected in L

$$\beta_{u,v}^{\alpha,\beta} = 0 \text{ if } L_{\alpha,\beta} = 0$$

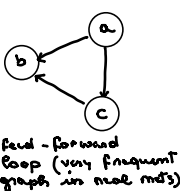
$$\sum_{u,v} \beta_{u,v}^{\alpha,\beta} > 0 \text{ if } L_{\alpha,\beta} = 1$$



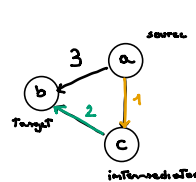
3 many systems are dynamical ⇒ RELATIONSHIPS EVOLVE

TIME for many networked-system is important INFO

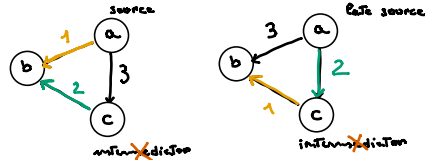
STATIC



feed-forward loop (very frequent graph in real net's)

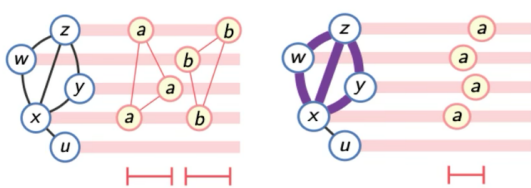


DYNAMIC

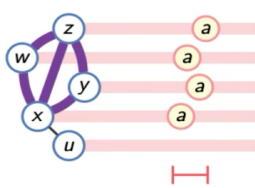


⇒ same static graph may be interpreted in 2 ways

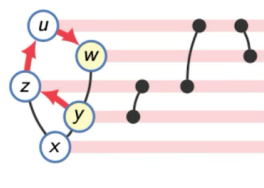
▷ SPECIFIC TASKS



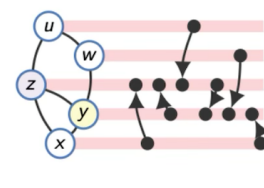
SYNCHRONIZATION OF SMALL SUBGROUPS
↓
BOTNET IDENTIFICATION



EVENTS
↓
burst of connected interactions



INFORMATION FLOW
Time can speedup or slow down information spreading



NETWORK EVOLUTION
how likely are interactions given a set of past interactions (temporally annotated)

the definition of temporal network is context-dependent, even "temporal network" may be the right term

Time	Relationships
Temporal	Graph
Dynamic	Network
Evolving	
Time-varying	
Time-dependent	
Evolutionary	

⇒ AT LEAST : format common to all data sources

$$I = \{(s, d, t, \Delta t)\}$$

when ↑ how long ↑
interacting nodes

▷ extensions : info application-dependent node lifespan

↓

TABULAR REPRESENTATION

A	B	1
A	B	2
B	C	3
C	D	4
C	D	5
B	C	5
B	D	6
C	D	7

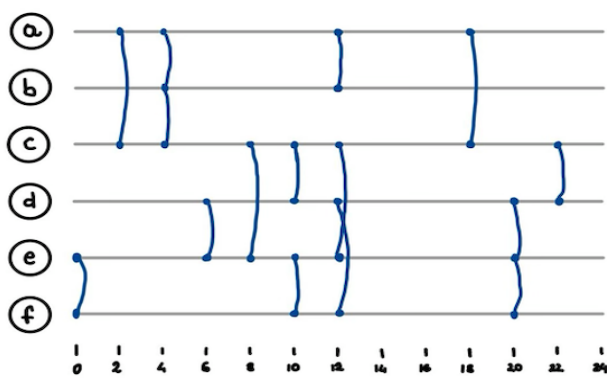
{(A, B, 1, 0),
(A, B, 2, 0),
(B, C, 3, 0),
(C, D, 4, 0),
(C, D, 5, 0),
(B, C, 5, 0),
(B, D, 6, 0),
(C, D, 7, 0)}

▷ each row is an interaction
▷ each column is an element of $(s, d, t, \Delta t)$
optional
at least 3 columns

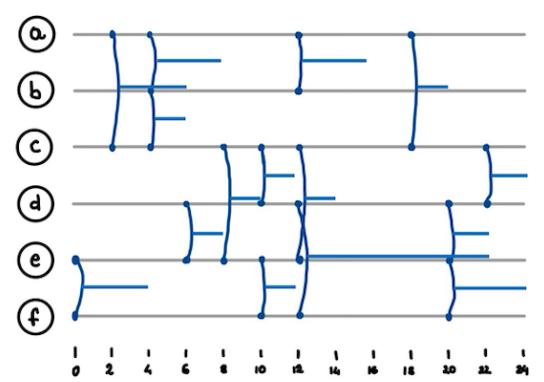
! "... when you see a paper, for example, temporal distance it could mean at least three different things, and look up the definitions paper by paper." P.Holme, 2015
! it's still this way

we can visualize interactions over time dropping Δt = instantaneous interactions or not drop

e	f	0	4
a	c	2	4
a	b	4	4
b	c	4	2
d	e	6	2
c	e	8	2
c	d	10	2
e	f	10	2
a	b	12	4
c	e	12	2
d	f	12	10
a	c	18	2
d	e	20	2
e	f	20	4
c	d	22	2



e	f	0	4
a	c	2	4
a	b	4	4
b	c	4	2
d	e	6	2
c	e	8	2
c	d	10	2
e	f	10	2
a	b	12	4
c	e	12	2
d	f	12	10
a	c	18	2
d	e	20	2
e	f	20	4
c	d	22	2



● STREAM GRAPH as TN model

$G = (T, V, W, E)$

simple and undirected

T = time domain (interval in $\mathbb{R}^+$ or in $\mathbb{N}^+$)

V = set of nodes

$W \subseteq T \times V$

$E \subseteq T \times V \times V$ s.t. $(t, \{u, v\}) \in E \Rightarrow (t, u) \in W, (t, v) \in W$
↓
unordered cartesian product

T_u = set of instant t when u is present

$\Rightarrow T_{uv} = \{t \mid \{u, v\} \text{ is active at } t\}$

if W is missing $\Rightarrow \forall u \in V, T_u = T \Rightarrow$ LINK STREAM MODEL

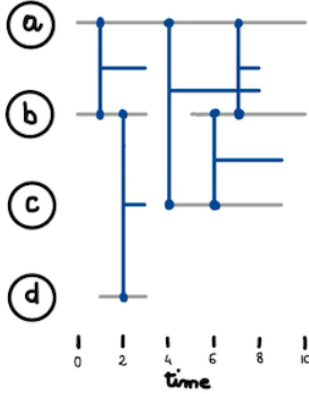
Here an example of stream graph

a	b	1	2
b	d	2	1
a	c	4	4
b	c	6	3
a	b	7	1

Interactions

a	0	10
b	0	3
b	5	5
c	4	5
d	1	2

Node life



$W = [0,10] \times \{a\} \cup ([0,3] \cup [5,10]) \times \{b\} \cup [4,9] \times \{c\} \cup [1,3] \times \{d\}$

$E = ([1,3] \cup [7,8]) \times \{ab\} \cup [2,3] \times \{bd\} \cup [4,8] \times \{ac\} \cup [6,9] \times \{bc\}$

$T_a = [0,1,2,3,4,5,6,7,8,9,10]$

$T_b = [0,1,2,3,5,6,7,8,9,10]$

$T_c = [5,6,7,8,9,10]$

$T_d = [1,2,3]$

$T_{ab} = [1,2,3] \cup [7,8]$

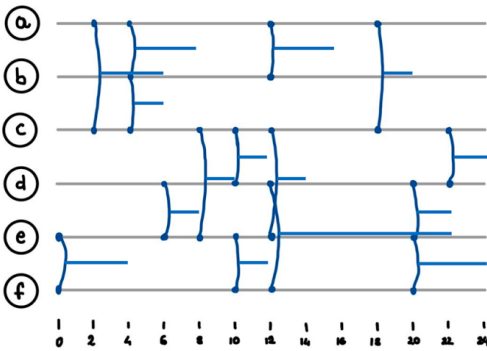
$T_{bd} = [2,3]$

$T_{ac} = [4,5,6,7,8]$

$T_{bc} = [6,7,8,9]$

and an example of link stream

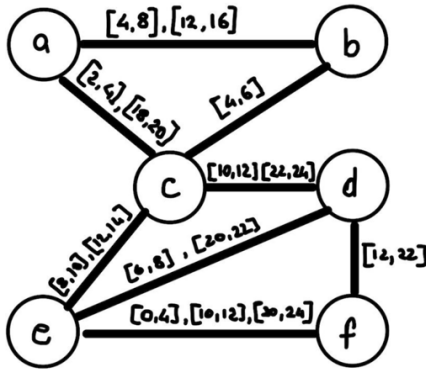
e	f	0	4
a	c	2	4
a	b	4	4
b	c	4	2
d	e	6	2
c	e	8	2
c	d	10	2
e	f	10	2
a	b	12	4
c	e	12	2
d	f	12	10
a	c	18	2
d	e	20	2
e	f	20	4
c	d	22	2



$\Rightarrow$ using link stream model as consequences on the implementation and representation
↓
we use edge-attribute graph notation where

$\tau: E \rightarrow \text{set of interval in } T \text{ when the link is active}$

INTERVAL GRAPH



● another notation equivalent to link stream is TIME-VARYING GRAPH

$G = (V, E, T, p, \lambda)$

T = time domain

V = set of nodes

$E \subseteq V \times V$

$p: E \times T \rightarrow \{0,1\}$ presence function

$\lambda: E \times T \rightarrow \mathbb{R}^+$ duration function

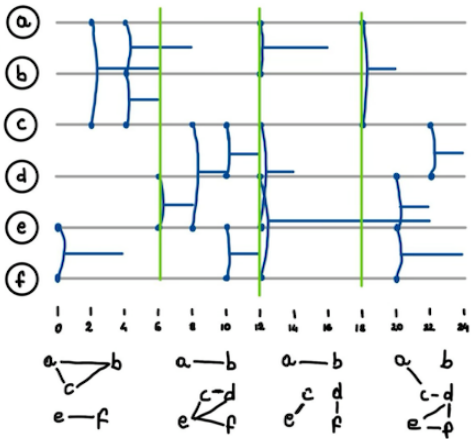
● a further representation is connected to multistage network $\Rightarrow$ SEQUENCE OF STATIC GRAPH

$\vec{G} = [G_1, \dots, G_{T_S}]$ where $G_i = (V_i, E_i)$

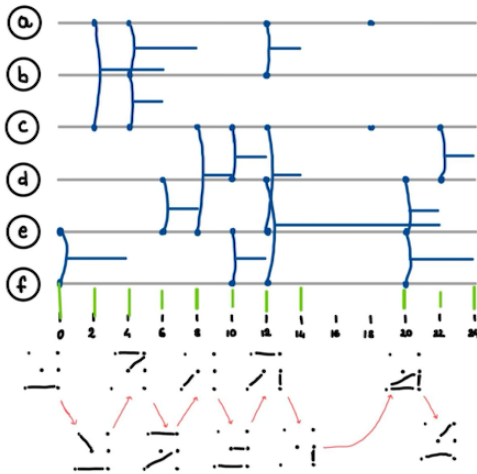
usually $V_i = V$ (nodes are fixed)

V_i = mode active in the time window i
 E_i = links " " " " " " i

TIME O SLIDING WINDOW is fundamental
↓
LOSSY $\approx$ LOSSLESS



Coarse Resolution



Variable Resolution

● lossiest representation TEMPORAL PROJECTED GRAPH

$\Rightarrow$ fix an interval $S = [a, b]$ and given temp. net.

$G_S = (V_S, E_S)$

$V_S = \{u, v \mid (u, v) \in E_S\}$

$E_S = \{(u, v) \mid S \cap T_{uv} \neq \emptyset\}$

$\Rightarrow$ extension : weighting functions on links $\approx$ duration, frequency